◉ DOMAIN 01 / AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

CLINIFLOW.

An end-to-end agentic clinical workflow for SME clinics — powered by an OpenAI-compatible LLM (ILMU GLM 5.1 in production) as the central reasoning core.

Dependency Free

Who am I ?

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Year 3 Software Engineering Student @ UM

Dr Lim SEES 40 PATIENTS A DAY

He spent more time *typing* than talking today.

Doctors spend **30–50%** of every shift on documentation

Patients repeat their history multiple times per visit

Records live across paper, Excel, and disconnected tools

Patients leave without understanding their treatment

THREE FATAL FLAWS

Every tool today *stops short* of the workflow.

FLAW · 01

Passive Note-Takers

"AI scribes" only transcribe. They don't reason, ask follow-ups, query patient history, or detect drug interactions.

▸ CliniFlow reasons, plans, and orchestrates — not just transcribes.

FLAW · 02

Hospital-Scale Only

Enterprise EMRs cost six figures and need IT teams. SME clinics — the majority of Malaysian primary care — get nothing built for them.

▸ Cloud SaaS · Docker Compose · 1-day deploy · no on-prem hardware.

FLAW · 03

No Patient Memory

Tools see each visit in isolation — no allergy lookup, no medication history, no drug-interaction check across past visits.

▸ Neo4j graph KB · multi-hop patient reasoning across every visit.

2 AGENTS · 3 PHASES · 1 LLM REASONING CORE · GRAPH KB

An *agentic workflow* for the entire visit.

Two specialised agents — **Pre-Visit Intake** and **Report** — orchestrated by an LLM. The Report Agent produces both the SOAP draft *and* the bilingual patient summary on finalize. A Neo4j graph KB carries patient context across visits.

PHASE 01 · PRE-VISIT

Intake Agent

Conversational symptom collection before arrival.

- Multi-turn ReAct loop
- Read-only tools · graph KB lookup
- Strict-JSON slot extraction
- Structured pre-visit report

PHASE 02 · VISIT

Report Agent

SOAP draft from voice or text · graph drug-interaction check.

- 3 input modes: live · upload · text
- Whisper STT (whisper-1)
- SOAP + medical entity extraction
- Adaptive rule learning loop

PHASE 03 · POST-VISIT

Finalize Tool

Bilingual patient summary — same agent, tool call.

- Plain-language EN + MS summary
- Structured medication guide
- Red-flag symptoms
- Patient portal delivery

01 · PRE-VISIT · INTAKE

Patient *talks first.* Doctor reads a brief.

A ReAct-loop agent runs the multi-turn intake. Tools are **read-only** — get_patient_context, get_visit_history — querying the Neo4j graph KB before asking. The agent decides when intake is complete, then a separate strict-JSON extract_slots() call produces the pre-visit report.

ReAct Loop | Tool Allowlist · Read-Only | Graph KB · Multi-Hop Query | Strict-JSON Slot Extraction | EN · MVP

CLINIFLOW · INTAKE
Hi! Before your visit with Dr. Lim — what's bothering you today?

PATIENT
I've had a bad cough for 3 days, sometimes a bit of blood.

CLINIFLOW · INTAKE
[graph KB · no penicillin allergy on file]
Cough for 3 days, with some blood. Any fever?

PATIENT
Yes, last night 38.5°C.

CLINIFLOW · INTAKE
Got it — intake complete. Generating your pre-visit report now.

▸ STRUCTURED REPORT GENERATED · DOCTOR NOTIFIED

02 · VISIT · REPORT AGENT

3 input modes. *One pipeline.*

Live recording (MediaRecorder), file upload (≤25 MB), or manual text — all converge on the same Report Agent. Whisper transcribes. The agent generates SOAP, runs a graph multi-hop drug interaction check, and persists the draft via update_soap_draft.

Live Record	File Upload	Manual Text
MEDIARECORDER	≤25MB · MP3/WAV	DEFAULT TAB

Whisper-1 STT · SOAP + ICD-10 Extract · Drug Interaction · Graph Multi-Hop · Adaptive Rule Injection

[S]	Productive cough × 3d, occasional haemoptysis. Fever 38.5°C since yesterday.
[O]	T: 38.4°C · BP 128/82 · SpO_2 96% · chest crackles RLL
[A]	Likely community-acquired pneumonia — RLL. R/O TB.
[P]	CXR · sputum AFB · amoxicillin 500mg TDS × 7d.
⚠ GRAPH	No allergy conflict · no active-med interaction.

▸ DRAFT PERSISTED · AWAITING DOCTOR FINALIZE

03 · POST-VISIT · FINALIZE

Same agent. *Tool call on finalize.*

Phase 3 is **not a separate agent** . When the doctor finalizes, the Report Agent calls a tool that produces the bilingual **EN + MS** patient summary — in the same DB transaction as the report write. Plain language, structured meds, red-flag symptoms.

Bilingual EN + MS | Same Transaction as Report Write | Red-Flag Symptom Extraction | Patient Portal Push

DIAGNOSIS	You have a chest infection (pneumonia). It's treatable.
MED · 01	Amoxicillin 500mg — 1 capsule, 3× daily, 7 days. With food.
MED · 02	Paracetamol 500mg — if fever >38°C. Max 4× daily.
RED FLAG	Difficulty breathing · chest pain · cough up more blood — go to ER.
FOLLOW-UP	Return in 3 days · or sooner if breathing worsens.

▸ DELIVERED EN + MS · CACHED IN MEDICAL_REPORTS

PLUGGABLE PROVIDER · OPENAI-COMPATIBLE · REACT LOOP

The LLM doesn't just generate. *It orchestrates.*

PROVIDER · PLUGGABLE

OpenAI-Compatible HTTP Client

```
# Dev: OpenAI · gpt-4o-mini GLM_BASE_URL = "https://api.openai.com/v1" GLM_MODEL    = "gpt-4o-mini"
# Prod: ILMU GLM 5.1 GLM_BASE_URL = "https://api.ilmu.ai/v1" GLM_MODEL    = "ilmu-glm-5.1" # STT:
separate from LLM provider OPENAI_BASE_URL = "https://api.openai.com/v1" STT_MODEL       = "whisper-
1"
```

REPORT AGENT · TOOL ALLOWLIST

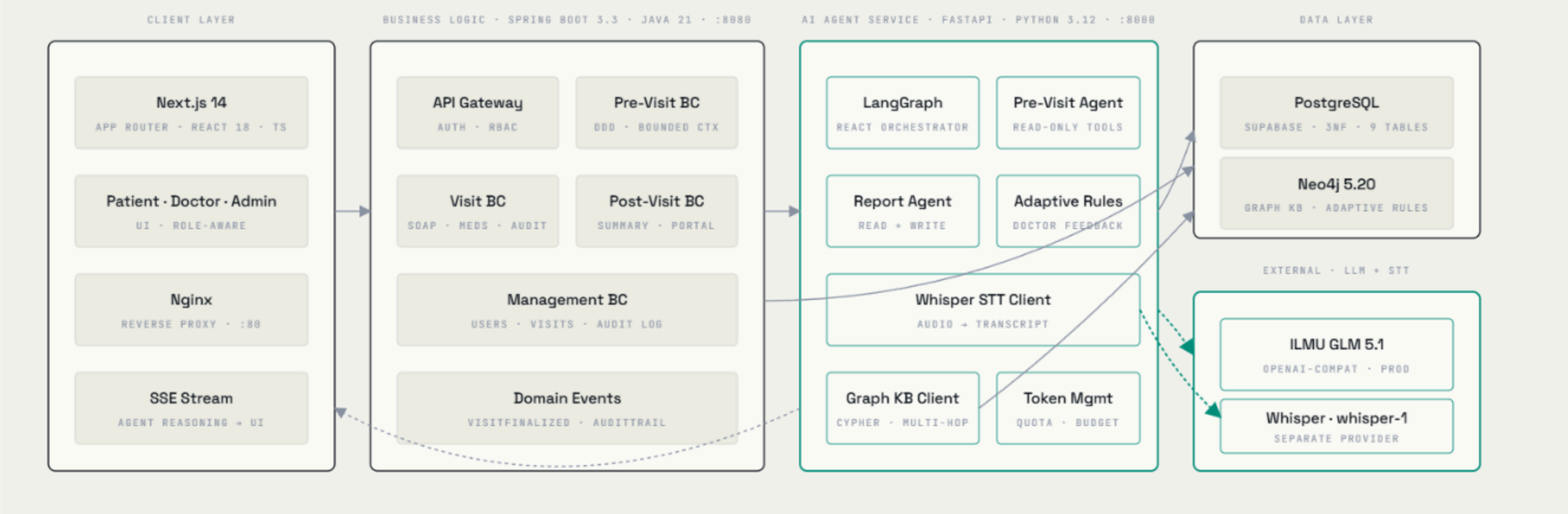
ReAct Loop · Tool-Calling

```
class ReportAgent (BaseAgent):
  agent_type =
"report"   tools = [
    get_visit_history,
# read PG + Neo4j     clinical_dictionary_extract, # ICD-10 sub-call     update_soap_draft,          # write PG
draft     record_inferred_edge,       # write Neo4j     emit_reasoning,            # SSE stream    ]

# HERMES_FENCE: rules ≠ diagnosis/dosing   # Doctor edits feed adaptive rule engine
```

NEXT.JS · SPRING BOOT · FASTAPI · POSTGRES + NEO4J

Polyglot by design. *DDD bounded contexts.*



- LLM tool calls · structured outputs
- REST + JDBC + Bolt protocol
- SSE stream · agent reasoning to UI
- VisitFinalized event · audit trail

SIX STAGES · ONE STATEFUL WORKFLOW · DOCTOR EDITS LEARN

INTAKE → CONSULT → SOAP → *REVIEW* → *FINALIZE* ↺

01
BOOTSTRAP

- Patient logs in
- Visit row created
- agent_turns init
- Audit begins

02
PRE-VISIT

- Intake Agent · ReAct
- Graph KB lookup
- Multi-turn loop
- extract_slots()

03
CONSULT

- 3 input modes
- Whisper STT
- Pre-visit brief shown
- Adaptive rules injected

04
STRUCTURE

- Report Agent · SOAP
- ICD-10 extract
- Drug interaction (graph)
- update_soap_draft

05
REVIEW

- Doctor edits inline
- Confidence flags shown
- Edit diff captured
- Adaptive rule proposed

06
FINALIZE

- Bilingual EN+MS summary
- Same DB transaction
- VisitFinalized event
- Patient portal push

Next.js 14	Spring Boot 3.3	FastAPI	LangGraph	PostgreSQL	Neo4j 5.20	ILMU GLM 5.1	Whisper-1
APP ROUTER · TS	JAVA 21 · DDD	PYTHON 3.12	REACT LOOP	SUPABASE · 3NF	GRAPH KB	LLM CORE · PROD	STT · OPENAI

Live Demo

REAL CLINICS ARE MESSY · WE DESIGNED FOR IT

Where most workflows *break* — CliniFlow degrades.

EDGE · 01

Low-confidence output

SOAP fields the LLM isn't sure about. Wrong dose or ambiguous diagnosis = clinical risk.

▸ report_confidence_flags jsonb · field-level · doctor must finalize.

EDGE · 02

LLM API outage

Cloud LLMs fail. The clinic can't stop seeing patients mid-shift.

▸ Manual-entry template · CRUD remains functional · queue for re-process.

EDGE · 03

Adaptive rule drift

A bad doctor edit could teach the system unsafe formatting.

▸ HERMES_FENCE · rules touch style only, never diagnosis or dosing.

EDGE · 04

Doctor disagrees with AI

AI must never override clinical judgment. The system must make this obvious.

▸ Draft never auto-promotes · explicit finalize · edit diff feeds learning.

MVP MODULES SHIPPED · POST-MVP MODULES PLANNED

Built for *depth*, not breadth.

Eight feature modules in the MVP — the smallest set that delivers the full agentic workflow end-to-end. Eight more modules are deliberately deferred, each with a real reason and a place on the future roadmap.

◉ MVP · MODULES SHIPPED

VHack 2026 build

- M1 **Identity & Access** — auth, RBAC, role-aware UI (patient/doctor/admin)
- M2 **Patient Records** — encrypted ID, demographics, visit history CRUD
- M3 **Pre-Visit Intake Agent** — ReAct loop · slot extraction · briefing
- M4 **Visit Capture** — 3 input modes: live record, file upload (≤25MB), text
- M5 **Report Agent & SOAP** — ICD-10 extract · confidence flags · doctor finalize
- M6 **Graph Knowledge Base** — Neo4j multi-hop · drug-interaction check
- M7 **Adaptive Rule Learning** — doctor edits → graph rules · HERMES_FENCE
- M8 **Bilingual Summary & Portal** — EN+MS finalize · patient portal · audit log

◉ POST-MVP · MODULES PLANNED

Deferred — with reasons

- F1 **BM Intake Module** — full Bahasa Malaysia chat (Phase 2 launch)
- F2 **Code-Switched STT** — mixed EN-MS audio (Whisper EN-only today)
- F3 **E-Prescribing Module** — deferred · Pharmacy Board compliance scope
- F4 **Telemedicine** — deferred · video infra is its own product
- F5 **Insurance Claims** — deferred · needs MY insurer integrations
- F6 **Medical Imaging** — deferred · outside workflow-automation scope
- F7 **Mobile Native App** — deferred · web-responsive prioritised first
- F8 **EHR-Grade Certification** — deferred · workflow first, full EHR later

UNDERSERVED · FRAGMENTED · READY

Hospitals get EHRs. *Clinics get nothing.*

◉ TARGET MARKET

PRIMARY

SME GP & specialist clinics — Malaysia

1–5 doctor practices · paper or basic digital today

SECONDARY

Community NGOs & corporate screening

High-volume · structured-record-poor environments

TERTIARY

ASEAN expansion — ID · VN · PH

Same SME profile · multi-lingual roadmap already planned

◉ PROPOSED REVENUE STREAMS

PRICING TO BE VALIDATED IN PILOT

01	Per-clinic SaaS subscription	**tiered**	BY VISIT VOLUME
02	Per-doctor seat add-on	**linear**	SCALES W/ TEAM
03	Deployment & training	**one-time**	SETUP · TRAINING

◉ GTM · KEY PARTNERS

MMA · clinic management software resellers · medical training schools · pilot through pre-existing clinic networks.

FIVE ADVANTAGES NO COMPETITOR COMBINES

Agentic. Graph-aware. *Doctor-safe.*

AI scribes only transcribe. Hospital EMRs are too heavy. CliniFlow is the only system built end-to-end as an agentic workflow with a graph KB, around a pluggable LLM, for Malaysian SME clinics.

EDGE · 01

True Agentic Workflow

2 ReAct agents · stateful · end-to-end. Not a chatbot bolted onto a form.

EDGE · 02

LLM at the Core

Pluggable OpenAI-compatible client. ILMU GLM 5.1 in prod. Provider can swap.

EDGE · 03

Graph Knowledge Base

Neo4j multi-hop · drug-interaction check · patient memory across visits.

EDGE · 04

Adaptive Rule Learning

Doctor edits → graph rules → injected into future prompts. The system learns.

EDGE · 05

Doctor-Safe by Design

HERMES_FENCE · doctor finalize required · append-only PDPA audit log.

FROM PRELIMINARY MVP TO VALIDATED LAUNCH

Build first. *Test. Then launch.*

We don't ship to clinics until the pilot validates the core hypothesis: ≥50% reduction in documentation time, with no clinically dangerous outputs. Each phase has a clear gate.

PHASE A · BUILD

VHACK 2026 → Q3 2026

Harden the MVP

- Finish all P0 features · 2 agents · 3 input modes
- Drug-interaction graph · adaptive rules
- PDPA hardening · encryption · RBAC · audit
- Internal QA suite · unit + integration tests
- **Gate:** All P0 acceptance criteria pass

PHASE B · TEST

Q4 2026 → Q2 2027

Klang Valley Pilot

- 3–5 SME clinics · supervised real-world use
- Canary → Beta → Majority rollout pattern
- A/B test adaptive rules vs baseline
- Doctor NPS > 30 · edit-rate ≥20% reduction
- **Gate:** ≥50% documentation-time reduction validated

PHASE C · LAUNCH

Q3 2027 →

Public Launch & Scale

- National Malaysia rollout · BM full release
- Clinic-management-software integrations
- ASEAN expansion — ID · VN · PH
- Toward EHR-grade certification · e-prescribing
- **Gate:** Pilot retention & clinical safety metrics held

◉ DEPENDENCY FREE · TANG SHAO XIAN

CLINI FLOW.

Doctors with patients, not paperwork. An agentic workflow built around the LLM as central reasoning core — exactly the way Domain 01 demands.

THANK YOU · QUESTIONS WELCOME